



**B.Sc. (General/Special) Degree
Programme
Faculty of Applied Sciences
University of Sri Jayewardenepura**

ICT 1022 2.0 Computer System Architecture

Presented By:

Surani Tissera(PhD)

Department of Computer Science

Introduction to the course



This is a property of Department of Computer Science, Faculty of Applied Science, University of Sri Jayewardenepura.

Course Outline

Course Name: ICT 1022 2.0 Computer System Architecture

Course type: Core course unit

Duration: 30 hours

Credits: 2.0



Aim of the course

- The aim of the course is to provide students with a solid foundation in understanding and designing modern computing systems.



Learning outcomes

At the completion of this course, You will be able to;

1. Describe the basic structure and functioning of computer systems.
2. Explain the principles of computer organization, design, and architecture.
3. Analyze performance metrics for computer systems and evaluate trade-offs.
4. Illustrate the working of processors, memory systems, and input/output modules.
5. Explain instruction set architectures and their implementation.



Assessment

- Formative Assessment (FA):
 - 30% of Total Marks from assignments
- Summative Assessment (SA):
 - 70% of Total Marks from End Semester Examination
 - Structured examination paper
- Attendance
 - **80% attendance is compulsory.**



Course Content

Module 1: Introduction to Computer System

Architecture

Module 2: Computer Organization Basics

Module 3: Data Path and Control

Module 4: Memory Systems

Module 5: Input/Output Systems



Reference

1. Patterson, D. A., & Hennessy, J. L. (2013). Computer Organization and Design: The Hardware/Software Interface.
2. Stallings, W. (2018). Computer Organization and Architecture: Designing for Performance.
3. Tanenbaum, A. S., & Austin, T. (2012). Structured Computer Organization.
4. Murdocca, M. J., & Heuring, V. P. (2007). Computer Architecture and Organization: An Integrated Approach.



EVOLUTION OF COMPUTER SYSTEM



This is a property of Department of Computer Science, Faculty of Applied Science, University of Sri Jayewardenepura.

Introduction

- Computers have become part and parcel of our daily lives.
 - They are everywhere (embedded systems?)
 - Laptops, tablets, mobile phones, intelligent appliances.
- It is required to understand how a computer works.
 - What are there inside a computer?
 - How does it work?
- We distinguish between two terms:

Computer Architecture and Computer Organization.



Computer Organization

- Design of the components and functional blocks using which computer systems are built.
- Analogy: civil engineer's task during building construction (cement, bricks, iron rods, and other building materials).

Computer Architecture:

- How to integrate the components to build a computer system to achieve a desired level of performance.
- Analogy: architect's task during the planning of a building (overall layout, floorplan, etc.).



Historical Perspective

Constant quest of building automatic computing machines have driven the development of computers.

- **Initial efforts:** mechanical devices like pulleys, levers and gears.
- **During World War II:** mechanical relays to carry out computations.
- **Vacuum tubes developed:** first electronic computer called ENIAC.
- **Semiconductor transistors developed and journey of miniaturization began.**

SSI -> MSI -> LSI -> VLSI -> ULSI.... Billions of transistors per chip



PASCALINE (1642)

- Mechanical calculator invented by B. Pascal.
- Could add and subtract two numbers directly, and multiply and divide by repetition.



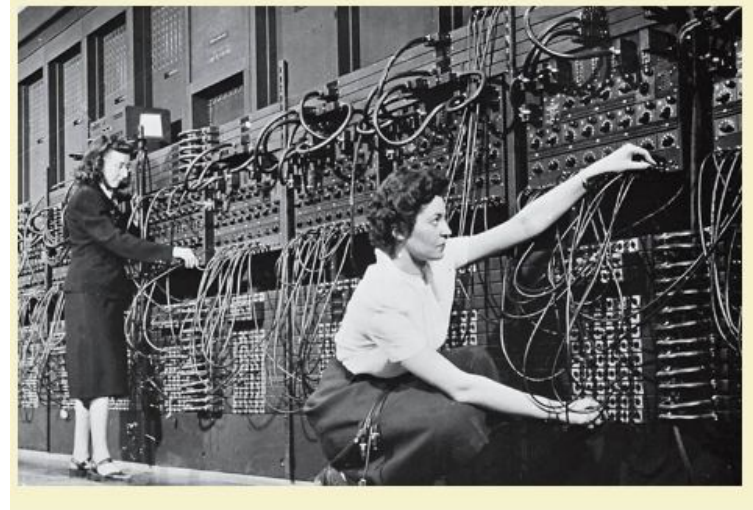
Babbage Engine

- First automatic computing engine was designed by Charles Babbage in the 19th century, but he could not build it.
- The first complete Babbage engine was built in 2002, 153 years later it was designed.
 - 8000 parts.
 - Weighed 5 tons.
 - 11 feet in length.



ENIAC

- Electrical Numerical Integrator and Calculator
- Developed at the University of Pennsylvania.
- Used 18,000 vacuum tubes, weighed 30 tons, and occupied a 30ft x 50ft space.



Harvard Mark 1

- Built at the University of Harvard in 1944, with support from IBM.
- Used mechanical relays (switches) to represent data.
- It weighed 35 tons, and required 500 miles of wiring.



IBM System/360

- Very popular mainframe computer of the 60s and 70s.
- Introduced many advanced architectural concepts that appeared in microprocessors several decades later.

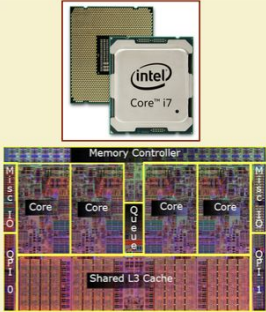


Intel Core i7

- A modern processor chip, that comes in dual-core, quad-core and 6-core variants.
- 64-bit processor that comes with various microarchitectures like Haswell, Nehalem, Sandy Bridge, etc.

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The diagram shows the internal architecture of an Intel Core i7 processor. It features four cores arranged in a 2x2 grid, each labeled 'Core'. Above the cores is the 'Memory Controller'. Below the cores is a 'Shared L3 Cache'. The diagram also shows various other components like 'QPI' and 'PCIe' interfaces.

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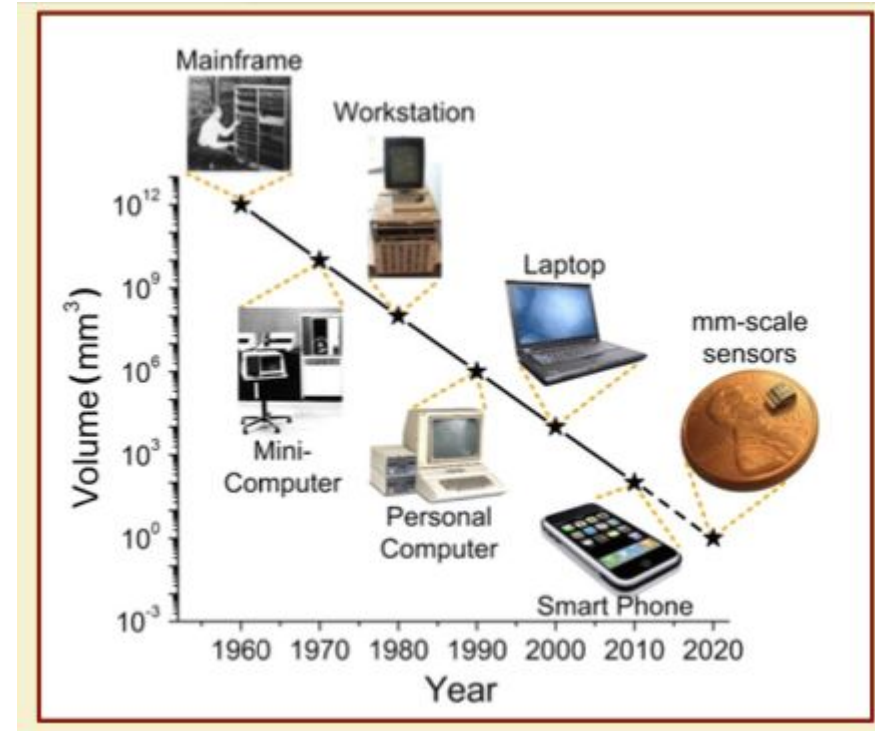
Generation	Main Technology	Representative Systems
First (1945-54)	Vacuum tubes, relays	Machine & assembly language ENIAC, IBM-701
Second (1955-64)	Transistors, memories, I/O processors	Batch processing systems, HLL IBM-7090
Third (1965-74)	SSI and MSI integrated circuits Microprogramming	Multiprogramming / Time sharing IBM 360, Intel 8008
Fourth (1975-84)	LSI and VLSI integrated circuits	Multiprocessors Intel 8086, 8088
Fifth (1984-90)	VLSI, multiprocessor on-chip	Parallel computing, Intel 486
Sixth (1990 onwards)	ULSI, scalable architecture, post-CMOS technologies	Massively parallel processors Pentium, SUN Ultra workstations



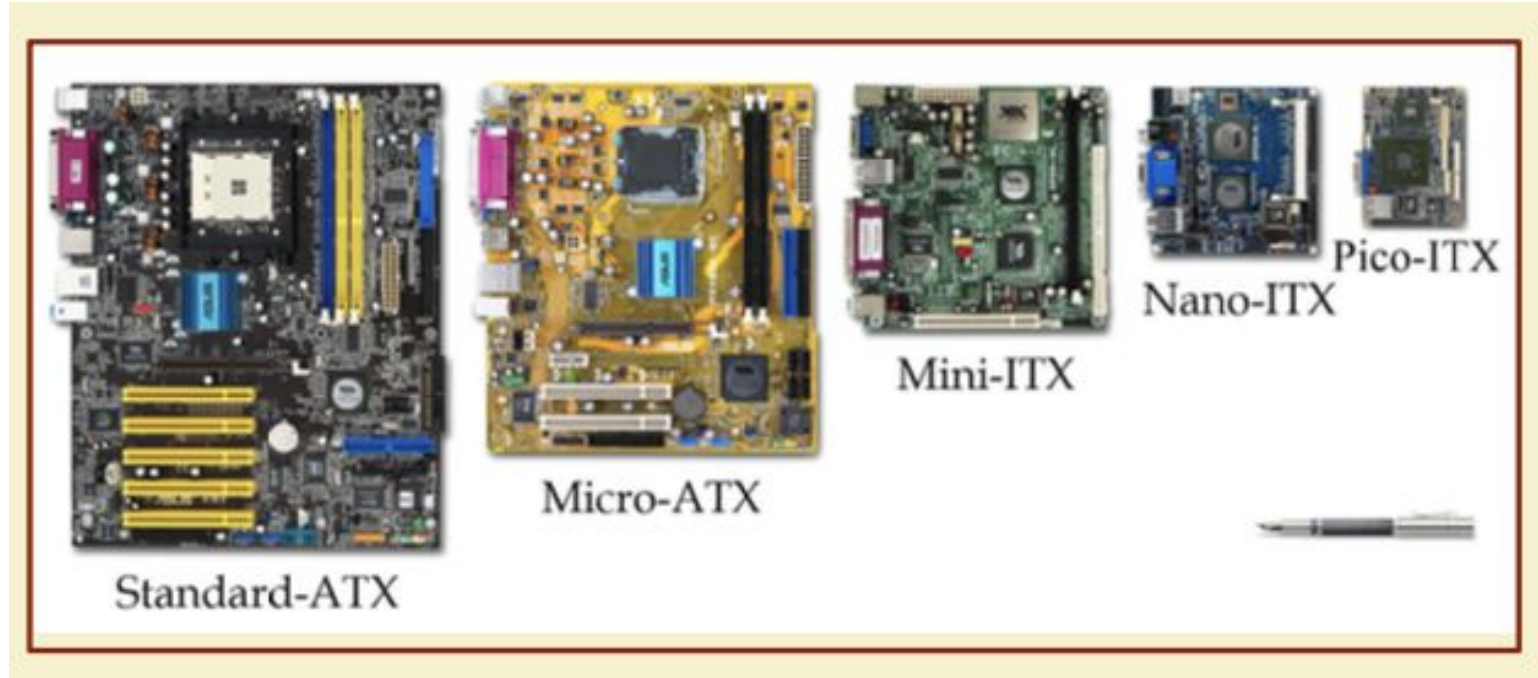
Evolution of the Types of Computer Systems

The future?

- Large-scale IoT based systems.
- Wearable computing.
- Intelligent objects.

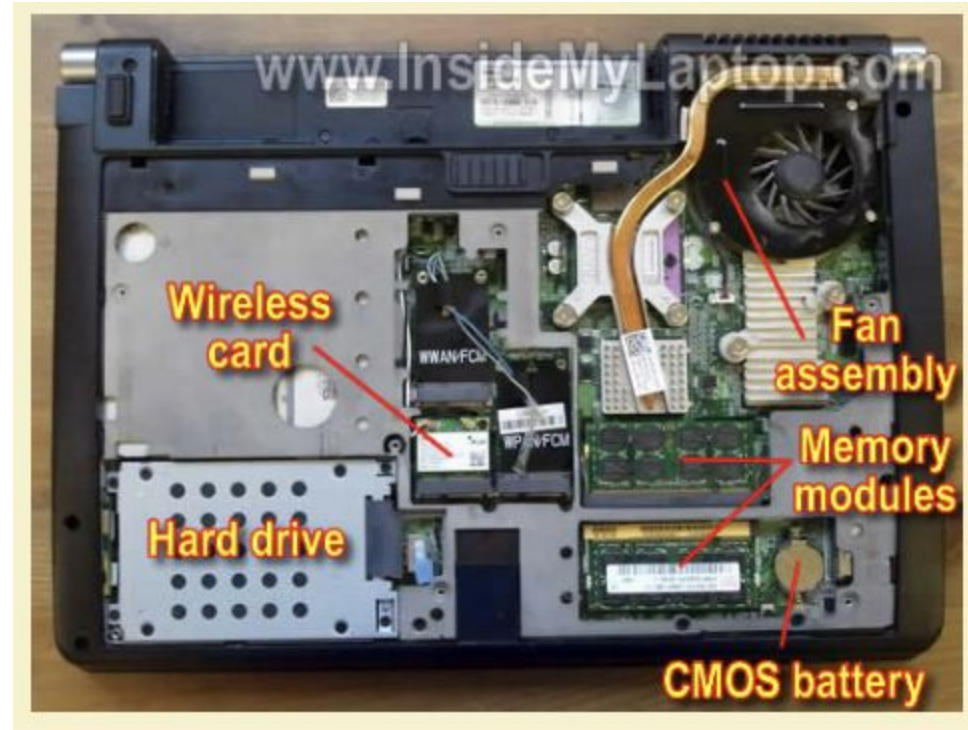


Evolution of PC form factors over the years



Inside a laptop

- Miniaturization in feature sizes of all parts.
- Hard drive being replaced by flash-based memory devices.
- Cooling is a major issue.

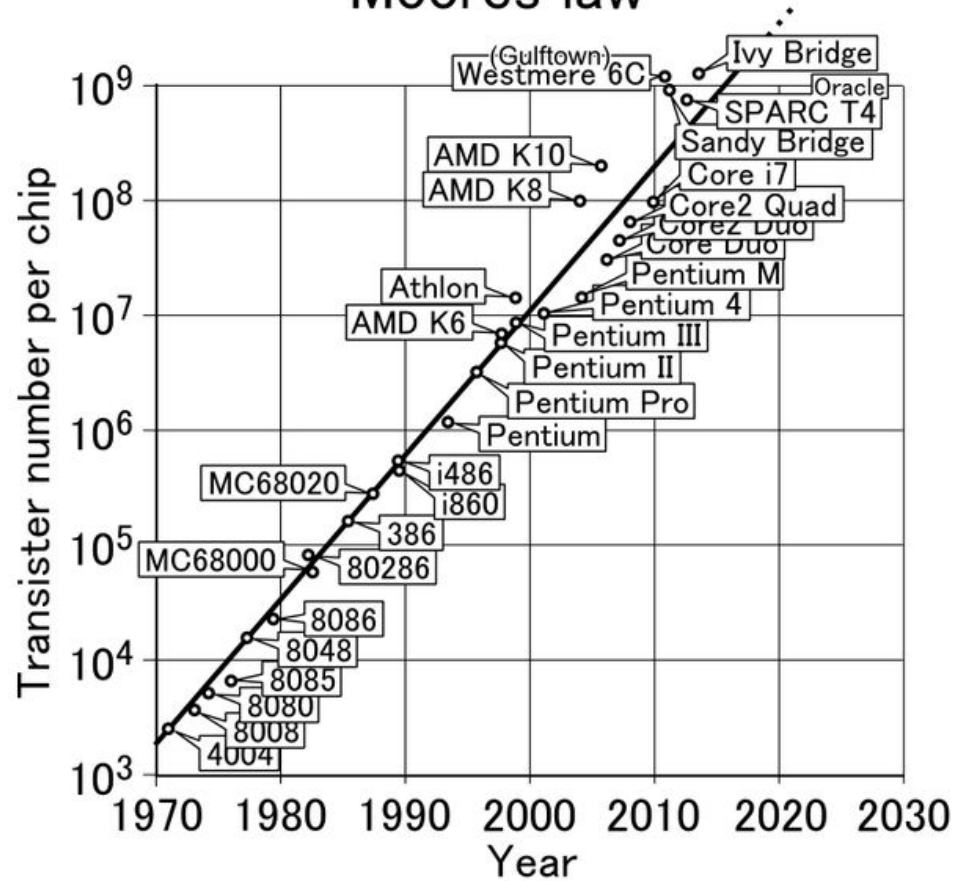


Moore's Law

- Refers to an observation made by Intel co-founder Gordon Moore in 1965. He noticed that the number of transistors per square inch on integrated circuits had doubled every year since their invention.
- Moore's law predicts that this trend will continue into the foreseeable future.
- Although the pace has slowed, the number of transistors per square inch has since doubled approximately every 18 months. This is used as the current definition of Moore's law.



Moore's law



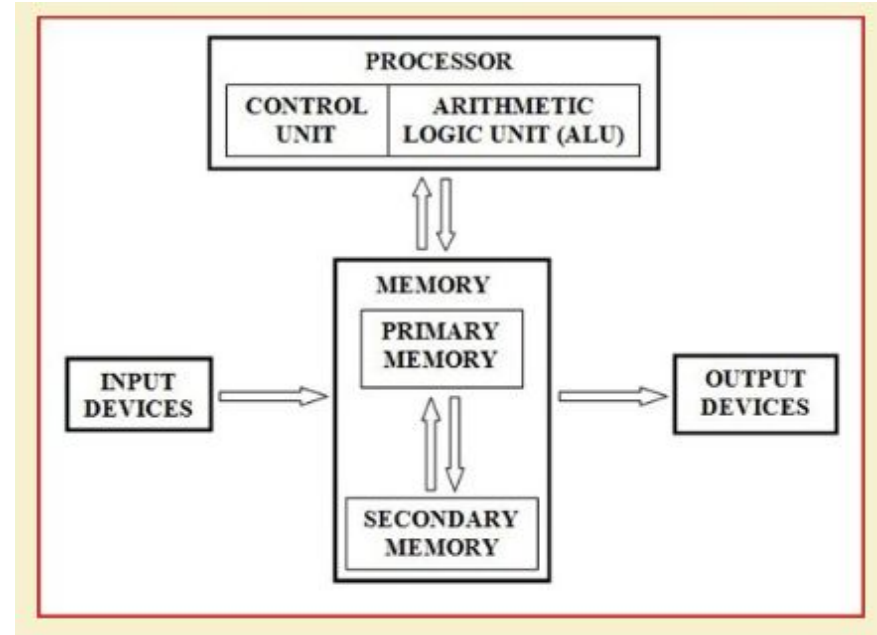
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Simplified Block Diagram of a Computer System

- All instructions and data are stored in memory.
- An instruction and the required data are brought into the processor for execution.
- Input and Output devices interface with the outside world.
- Referred to as **von-Neumann architecture**.



Inside the Processor

- Also called Central Processing Unit (CPU).
- Consists of a Control Unit and an Arithmetic Logic Unit (ALU).
- All calculations happen inside the ALU.
- The Control Unit generates sequence of control signals to carry out all operations.
- The processor fetches an instruction from memory for execution.
- An instruction specifies the exact operation to be carried out.
- It also specifies the data that are to be operated on.
- A program refers to a set of instructions that are required to carry out some specific task (e.g. sorting a set of numbers).



What is the role of ALU?

- It contains several registers, some general-purpose and some special-purpose, for temporary storage of data.
- It contains circuitry to carry out logic operations, like AND, OR, NOT, shift, compare, etc.
- It contains circuitry to carry out arithmetic operations like addition, subtraction, multiplication, division, etc.
- During instruction execution, the data (operands) are brought in and
- stored in some registers, the desired operation carried out, and the result stored back in some register or memory.



What is the role of control unit?

- Acts as the nerve center that senses the states of various functional units and sends control signals to control their states.
- To carry out a specific operation (say, $R1 \rightarrow R2 + R3$), the control unit must generate control signals in a specific sequence.
 - Enable the outputs of registers R2 and R3.
 - Select the addition operation.
 - Store the output of the adder circuit into register R1.
- When an instruction is fetched from memory, the operation (called opcode) is decoded by the control unit, and the control signals issued.



Inside the Memory Unit

- Two main types of memory subsystems.
 - Primary or Main memory, which stores the active instructions and data for the program being executed on the processor.
 - Secondary memory, which is used as a backup and stores all active and inactive programs and data, typically as files.
- The processor only has direct access to the primary memory.
- In reality, the memory system is implemented as a hierarchy of several levels.
 - L1 cache, L2 cache, L3 cache, primary memory, secondary memory.
 - Objective is to provide faster memory access at affordable cost.



Various different types of memory are possible.

- a) **Random Access Memory (RAM)**, which is used for the cache and primary memory subsystems. Read and Write access times are independent of the location being accessed.
- b) **Read Only Memory (ROM)**, which is used as part of the primary memory to store some fixed data that cannot be changed.
- c) **Magnetic Disk**, which uses direction of magnetization of tiny magnetic particles on a metallic surface to store data. Access times vary depending on the location being accessed, and is used as secondary memory.
- d) **Flash Memory**, which is replacing magnetic disks as secondary memory devices. They are faster, but smaller in size as compared





Input Unit

- Used to feed data to the computer system from the external environment.
 - Data are transferred to the processor/memory after appropriate encoding.
- Common input devices:
 - Keyboard
 - Mouse
 - Joystick
 - Camera



Input Unit



Output **Unit**

- Used to send the result of some computation to the outside world.
- Common output devices:
 - LCD/LED screen
 - Printer and Plotter
 - Speaker / Buzzer
 - Projection system



Output Unit



Thank
You!

